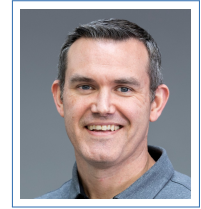


# Christian Bird

*Senior Principal Researcher*

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## Research Interests

Dr. Christian Bird is a Senior Principal Researcher at Microsoft Research, where he studies software engineering practices, developer productivity, and the intersection of AI and software development. His work combines empirical software engineering, large-scale data analysis, and human factors to understand how developers work and how tools can better support them.

## Education

- 2010 **Ph.D. Computer Science**, *University of California, Davis, California*  
Advisor: Premkumar Devanbu  
Thesis: "Sociotechnical Collaboration and Coordination in Open Source Software"
- 2008 **M.S. Computer Science**, *University of California, Davis, California*  
Advisor: Premkumar Devanbu
- 2003 **B.S. Computer Science**, *Brigham Young University, Provo, Utah*

## Honors and Awards

- Distinguished Scientist**, *Association for Computing Machinery (ACM)*, 2024
- ACM SIGSOFT Impact Paper Award (11+ years)**, *Awarded at ICSE 2025, "Expectations, Outcomes, and Challenges of Modern Code Review"*, 2025
- Test of Time Honorable Mention (10 years)**, *ESEC/FSE, "Suggesting Accurate Method and Class Names"*, 2025
- Test of Time Award (10 years)**, *ESEC/FSE, "Don't Touch My Code! Examining the Effects of Ownership on Software Quality"*, 2021
- 30 Year Highlight (26 most influential papers in the past 30 years)**, *ISSRE, "Putting It All Together: Using Socio-Technical Networks to Predict Failures"*, 2019
- Test of Time Award (10 years)**, *ESEC/FSE, "Fair and balanced?: bias in bug-fix datasets"*, 2019
- Most Influential Paper Award (10 years)**, *International Conference on Mining Software Repositories, "The Promises and Perils of Mining Git"*, 2019

**IEEE Software ICSE Software Engineering in Practice Best Paper Award**, *International Conference on Software Engineering*, "Software Engineering for Machine Learning: A Case Study", 2019

**Distinguished Reviewer**, *International Conference on Software Engineering*, 2019

**Test of Time Award (10 years)**, *ESEC/FSE*, "Latent Social Structure in Open Source Projects", 2018

**ACM SIGSOFT Early Career Research Award**, *Awarded at ICSE 2017*

**Most Influential Paper Award (10 years)**, *International Conference on Mining Software Repositories*, "Mining Email Social Networks", 2016

**Distinguished Reviewer**, *International Conference on Software Engineering*, 2016

**Distinguished Reviewer**, *IEEE/ACM International Conference on Automated Software Engineering*, 2015

**ACM SIGSOFT Distinguished Paper**, *ACM SigSoft Symposium on Foundations of Software Engineering*, "Learning Natural Coding Conventions", 2014

**ACM SIGSOFT Distinguished Paper and ISSTA Best Paper**, *ACM SigSoft International Symposium on Software Testing and Analysis*, "Collecting a Heap of Shapes", 2013

**Distinguished Referee**, *ACM Transactions on Software Engineering and Methodology*, 2011-2012

**ACM SIGSOFT Distinguished Paper**, *ACM SigSoft Symposium on Foundations of Software Engineering*, "Assessing the Value of Branches with What-If Analysis", 2012

**ACM SIGSOFT Distinguished Paper**, *International Conference on Software Engineering*, "Does Distributed Development Affect Software Quality? An Empirical Case Study of Windows Vista", 2009

**Mining Software Repositories Best Paper Award**, *International Working Conference on Mining Software Repositories*, "Clones: What is that smell?", 2010

**Best Student Talk**, *Mining Software Archives*, "The Effect of Change Semantics on Coordination Requirements", 2010

**Best Graduate Researcher Award**, *U.C. Davis, Dept. of Computer Science*, 2008 – 2009

**Dean's List**, *Brigham Young University*, 1996, 2001, 2002

**US National Merit Scholar and Scholarship**, Awarded for complete undergraduate career, 1996

**Boy Scouts of America: Eagle Scout Rank**, 1996

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## Books

Christian Bird, Tim Menzies, and Thomas Zimmermann, Eds. *The Art and Science of Analyzing Software Data*. Morgan Kauffman, 2015

## Special Issues Edited

**Special Issue on Mining Software Repositories 2015**, *Journal of Empirical Software Engineering*, 2018

**Special Issue on Release Engineering 3.0**, *IEEE Software*, 2018

**Special Issue on Program Comprehension 2014**, *Journal of Empirical Software Engineering*, 2017

**Special Issue on Crowdsourcing**, *IEEE Software*, 2017

**Special Issue on Release Engineering**, *IEEE Software*, 2015

## Book Chapters

Christian Bird. Interviews. In *Perspectives on Data Science for Software Engineering*, T. Menzies, L. Williams, and T. Zimmermann, Eds. Morgan Kaufmann, 2016

Christian Bird. Don't embarrass yourself: Beware of bias in your data. In *Perspectives on Data Science for Software Engineering*, T. Menzies, L. Williams, and T. Zimmermann, Eds. Morgan Kaufmann, 2016

Christian Bird. Conway's Corollary. In *Making Software: What Really Works, and Why We Believe It*, A. Oram and G. Wilson, Eds. O'Reilly, 2010

## Refereed Publications

- TOSEM 2026 Brittany Johnson, Christian Bird, Denae Ford, Ebtesam Al Haque, Nicole Forsgren, and Thomas Zimmermann. Facilitating Trust in AI-assisted Software Tools. *ACM Transactions on Software Engineering and Methodology (TOSEM)*, 2026. To appear
- ICSE 2025 Nadia Nahar, Christian Kästner, Jenna L. Butler, Chris Parnin, Thomas Zimmermann, and Christian Bird. Beyond the Comfort Zone: Emerging Solutions to Overcome Challenges in Integrating LLMs into Software Products. In *47th IEEE/ACM International Conference on Software Engineering: Software Engineering in Practice Track*, 2025, IEEE, pp. 516–527
- ICSE 2025 Ajay Krishna Vajjala, Arun Krishna Vajjala, Carmen Badea, Christian Bird, Jade D'Souza, Robert DeLine, Mikhail O. Demyanyuk, Jason Entenmann, Nicole Forsgren, Aliaksandr Hramadski, Haris Mohammad, Sandeepan Sanyal, Oleg Surmachev, and Thomas Zimmermann. Enhancing Differential Testing: LLM-Powered Automation in Release Engineering. In *47th IEEE/ACM International Conference on Software Engineering: Software Engineering in Practice Track*, 2025, IEEE, pp. 607–617

- FSE 2025 Arun Krishna Vajjala, Ajay Krishna Vajjala, Carmen Badea, Christian Bird, Jade D’Souza, Robert DeLine, Mikhail O. Demyanyuk, Jason Entenmann, Nicole Forsgren, Aliaksandr Hramadski, Haris Mohammad, Sandeepan Sanyal, Oleg Surmachev, and Thomas Zimmermann. Using Large Language Models to Support the Workflow of Differential Testing. In *Proceedings of the 33rd ACM International Conference on the Foundations of Software Engineering*, 2025, L. Montecchi, J. Li, D. Poshyvanyk, and D. Zhang, Eds., ACM, pp. 355–365
- CACM 2025 Shuvendu K. Lahiri, Alexey Svyatkovskiy, Christian Bird, Erik Miejer, and Terry Coatta. Program Merge: What’s Deep Learning Got to Do with It? *Communications of the ACM* 68, 3, 2025, 34–41
- ICSE 2024 Jenna L. Butler, Thomas Zimmermann, and Christian Bird. Objectives and Key Results in Software Teams: Challenges, Opportunities and Impact on Development. In *Proceedings of the 46th International Conference on Software Engineering: Software Engineering in Practice Track*, 2024, ACM, pp. 358–368
- FSE 2024 Jenny T. Liang, Carmen Badea, Christian Bird, Robert DeLine, Denae Ford, Nicole Forsgren, and Thomas Zimmermann. Can GPT-4 Replicate Empirical Software Engineering Research? vol. 1, pp. 1330–1353
- ACM Queue 2024 Shuvendu K. Lahiri, Alexey Svyatkovskiy, Christian Bird, Erik Meijer, and Terry Coatta. Program Merge: What’s Deep Learning Got to Do with It?: A discussion with Shuvendu Lahiri, Alexey Svyatkovskiy, Christian Bird, Erik Meijer and Terry Coatta. *ACM Queue* 22, 4, 2024, 50
- ICSE 2023 Brittany Johnson, Christian Bird, Denae Ford, Nicole Forsgren, and Thomas Zimmermann. Make Your Tools Sparkle with Trust: The PICSE Framework for Trust in Software Tools. In *45th IEEE/ACM International Conference on Software Engineering: Software Engineering in Practice Track*, 2023, IEEE, pp. 409–419
- ICSE 2023 Jiyang Zhang, Chandra Shekhar Maddila, Ram Bairi, Christian Bird, Ujjwal Raizada, Apoorva Agrawal, Yamini Jhawar, Kim Herzig, and Arie van Deursen. Using Large-scale Heterogeneous Graph Representation Learning for Code Review Recommendations at Microsoft. In *45th IEEE/ACM International Conference on Software Engineering: Software Engineering in Practice Track*, 2023, IEEE, pp. 162–172
- CACM 2023 Christian Bird, Denae Ford, Thomas Zimmermann, Nicole Forsgren, Eirini Kalliamvakou, Travis Lowdermilk, and Idan Gazit. Taking Flight with Copilot. *Communications of the ACM* 66, 6, 2023, 56–62
- TSE 2023 Elizabeth Dinella, Todd Mytkowicz, Alexey Svyatkovskiy, Christian Bird, Mayur Naik, and Shuvendu K. Lahiri. DeepMerge: Learning to Merge Programs. *IEEE Transactions on Software Engineering* 49, 4, 2023, 1599–1614

- FSE 2022 Alexey Svyatkovskiy, Sarah Fakhoury, Negar Ghorbani, Todd Mytkowicz, Elizabeth Dinella, Christian Bird, Jinu Jang, Neel Sundaresan, and Shuvendu K. Lahiri. Program merge conflict resolution via neural transformers. In *Proceedings of the 30th ACM Joint European Software Engineering Conference and Symposium on the Foundations of Software Engineering*, 2022, A. Roychoudhury, C. Cadar, and M. Kim, Eds., ACM, pp. 822–833
- HCI 2022 Ei Pa Pa Pe-Than, Alexander Nolte, Anna Filippova, Christian Bird, Steve Scallen, and James D. Herbsleb. Corporate hackathons, how and why? A multiple case study of motivation, projects proposal and selection, goal setting, coordination, and outcomes. *Journal of Human-Computer Interaction* 37, 4, 2022, 281–313
- ACM Queue 2022 Christian Bird, Denae Ford, Thomas Zimmermann, Nicole Forsgren, Eirini Kalliamvakou, Travis Lowdermilk, and Idan Gazit. Taking Flight with Copilot: Early insights and opportunities of AI-powered pair-programming tools. *ACM Queue* 20, 6, 2022, 35–57
- TOSEM 2022 Denae Ford, Margaret-Anne D. Storey, Thomas Zimmermann, Christian Bird, Sonia Jaffe, Chandra Shekhar Maddila, Jenna L. Butler, Brian Houck, and Nachiappan Nagappan. A Tale of Two Cities: Software Developers Working from Home during the COVID-19 Pandemic. *ACM Transactions on Software Engineering and Methodology* 31, 2, 2022, 27:1–27:37
- TOSEM 2022 Chandra Shekhar Maddila, Nachiappan Nagappan, Christian Bird, Georgios Gousios, and Arie van Deursen. ConE: A Concurrent Edit Detection Tool for Large-scale Software Development. *ACM Transactions on Software Engineering and Methodology* 31, 2, 2022, 22:1–22:26
- TSE 2021 Margaret-Anne D. Storey, Thomas Zimmermann, Christian Bird, Jacek Czerwinka, Brendan Murphy, and Eirini Kalliamvakou. Towards a Theory of Software Developer Job Satisfaction and Perceived Productivity. *IEEE Transactions on Software Engineering* 47, 10, 2021, 2125–2142
- TSE 2021 Pavneet Singh Kochhar, Eirini Kalliamvakou, Nachiappan Nagappan, Thomas Zimmermann, and Christian Bird. Moving from Closed to Open Source: Observations from Six Transitioned Projects to GitHub. *IEEE Transactions on Software Engineering* 47, 9, 2021, 1838–1856
- TSE 2021 André N. Meyer, Earl T. Barr, Christian Bird, and Thomas Zimmermann. Today Was a Good Day: The Daily Life of Software Developers. *IEEE Transactions on Software Engineering* 47, 5, 2021, 863–880
- TSE 2021 Brittany Johnson, Thomas Zimmermann, and Christian Bird. The Effect of Work Environments on Productivity and Satisfaction of Software Engineers. *IEEE Transactions on Software Engineering* 47, 4, 2021, 736–757
- ICSE 2020 Jordan Henkel, Christian Bird, Shuvendu K. Lahiri, and Thomas Reps. Learning from, Understanding, and Supporting DevOps Artifacts for Docker. In *Proceedings of the International Conference on Software Engineering*, 2020

- MSR 2020 Jordan Henkel, Christian Bird, Shuvendu K. Lahiri, and Thomas Reps. A Dataset of Dockerfiles. In *Proceedings of the International Conference on Mining Software Repositories*, 2020
- NSDI 2020 Sonu Mehta, Ranjita Bhagwan, Rahul Kumar, Chetan Bansal, Chandra Shekhar Maddila, B. Ashok, Sumit Asthana, Christian Bird, and Aditya Kumar. Rex: Preventing bugs and misconfiguration in large services using correlated change analysis. In *17th USENIX Symposium on Networked Systems Design and Implementation, NSDI 2020, Santa Clara, CA, USA, February 25-27, 2020*, 2020, USENIX Association, pp. 435–448
- FSE 2019 Sumit Asthana, Rahul Kumar, Ranjita Bhagwan, Christian Bird, Chetan Bansal, Chandra Shekhar Maddila, Sonu Mehta, and B. Ashok. Whodo: automating reviewer suggestions at scale. In *Proceedings of the ACM Joint Meeting on European Software Engineering Conference and Symposium on the Foundations of Software Engineering (Industry Track)*, 2019, pp. 937–945
- ICSE 2019 Saleema Amershi, Andrew Begel, Christian Bird, Robert DeLine, Harald Gall, Ece Kamar, Nachiappan Nagappan, Besmira Nushi, and Thomas Zimmermann. Software Engineering for Machine Learning: A Case Study. In *Proceedings of the International Conference on Software Engineering (Software Engineering In Practice Track)*, 2019, IEEE
- TSE 2019 Vladimir Kovalenko, Nava Tintarev, Evgeny Pasyukov, Christian Bird, and Alberto Bacchelli. Does Reviewer Recommendation Help Developers? *IEEE Transactions on Software Engineering*, to appear 2019
- CHI 2018 Alexander Nolte, Ei Pa Pa Pe-Than, Anna Filippova, Christian Bird, Steve Scallen, and James D Herbsleb. You Hacked and Now What?:-Exploring Outcomes of a Corporate Hackathon. *Proceedings of the ACM on Human-Computer Interaction 2*, CSCW, 2018, 129
- ACM Queue 2018 Jacek Czerwinka, Michaela Greiler, Christian Bird, Lucas Panjer, and Terry Coatta. Codeflow: Improving the code review process at microsoft. *ACM Queue* 16, 5, 2018, 20
- IEEE Software 2018 Ei Pa Pa Pe Than, Alexander Nolte, Anna Filippova, Christian Bird, Steve Scallen, and James Herbsleb. Designing Corporate Hackathons With a Purpose. *IEEE Software*, Early Access 2018
- FSE 2018 Vincent J Hellendoorn, Christian Bird, Earl T Barr, and Miltiadis Allamanis. Deep learning type inference. In *Proceedings of the 2018 26th ACM Joint Meeting on European Software Engineering Conference and Symposium on the Foundations of Software Engineering*, 2018, ACM, pp. 152–162
- IEEE Software 2018 Laura Macleod, Michaela Greiler, Margaret-Anne Storey, Christian Bird, and Jacek Czerwinka. Code Reviewing in the Trenches: Understanding Challenges, Best Practices, and Tool Needs. *IEEE Software*, 2017
- TSE 2018 Miltiadis Allamanis, Earl Barr, Christian Bird, Premkumar Devanbu, Mark Marron, and Charles Sutton. Mining Semantic Loop Idioms from Big Code. *IEEE Transactions on Software Engineering*, 2018

- CHI 2018 Austin Z Henley, Kivanç Muçlu, Maria Christakis, Scott D Fleming, and Christian Bird. Cfar: A tool to increase communication, productivity, and review quality in collaborative code reviews. In *Proceedings of the 2018 CHI Conference on Human Factors in Computing Systems*, 2018, ACM, p. 157
- TSE 2018 Eirini Kalliamvakou, Christian Bird, Thomas Zimmermann, Andrew Begel, Robert DeLine, and Daniel M. German. What Makes a Great Manager of Software Engineers? *IEEE Transactions on Software Engineering*, 2018
- ESEM 2017 Denae Ford, Thomas Zimmermann, Christian Bird, and Nachiappan Nagappan. Characterizing Software Engineering Work with Personas Based on Knowledge Worker Actions. In *Proceedings of the International Symposium on Empirical Software Engineering and Measurement*, 2017, ACM/IEEE
- ICSE 2017 Zheng Gao, Christian Bird, and Earl T. Barr. To Type or not to Type: On the Effectiveness of Static typing for JavaScript. In *Proceedings of the 39th International Conference on Software Engineering*, 2017, IEEE. (16%)
- TSE 2017 Amiangshu Bosu, Jeffrey C. Carver, Christian Bird, Jonathan Orbeck, and Chris Chockley. Process Aspects and Social Dynamics of Contemporary Code Review: Insights from Open Source Development and Industrial Practice at Microsoft. *IEEE Transactions on Software Engineering*, 2017
- ASE 2016 Maria Christakis, and Christian Bird. What developers want and need from program analysis: an empirical study. In *Proceedings of the 31st IEEE/ACM International Conference on Automated Software Engineering*, 2016, ACM, pp. 332–343
- ICSE 2016 Irene Manotas, Christian Bird, Rui Zhang, David Shepherd, Will Snipes, Ciera Jaspan, Caitlin Sadowski, Lori Pollock, and James Clause. An Empirical Study of Practitioners' Perspectives on Green Software Engineering. In *Proceedings of the 38th International Conference on Software Engineering*, 2016, ACM. (19%)
- ICSE 2016 Premkumar Devanbu, Thomas Zimmermann, and Christian Bird. Belief and Evidence in Empirical Software Engineering. In *Proceedings of the 38th International Conference on Software Engineering*, 2016, ACM. (19%)
- ICSE 2016 Michael Washburn, Pvithra Sathiyarayanan, Meiyappan Nagappan, Thomas Zimmermann, and Christian Bird. What Went Right and What Went Wrong: An Analysis of 155 Postmortems from Game Development. In *Proceedings of the 38th International Conference on Software Engineering (Software Engineering in Practice Track)*, 2016, ACM. (26%)
- CHASE 2016 Edward K Smith, Christian Bird, and Thomas Zimmermann. Beliefs, practices, and personalities of software engineers: a survey in a large software company. In *Proceedings of the 9th International Workshop on Cooperative and Human Aspects of Software Engineering*, 2016, ACM, pp. 15–18
- TSE 2016 Motahareh Bahrami Zanjani, Huzefa Kagdi, and Christian Bird. Automatically Recommending Peer Reviewers in Modern Code Review. *IEEE Transactions on Software Engineering* 42, 6, June 2016, 530–543

- FSE 2015 Miltiadis Allamanis, Earl T. Barr, Christian Bird, and Charles Sutton. Suggesting Accurate Method and Class Names. In *Proceedings of the the joint meeting of the European Software Engineering Conference and the ACM SIGSOFT Symposium on The Foundations of Software Engineering (ESEC/FSE)*, 2015, ACM. (25%)
- FSE 2015 Juliana Saraiva, Christian Bird, and Thomas Zimmermann. Products, Developers, and Milestones: How Should I Build My N-Gram Language Model. In *Proceedings of the the joint meeting of the European Software Engineering Conference and the ACM SIGSOFT Symposium on The Foundations of Software Engineering (ESEC/FSE) Industry Track*, 2015, ACM. (33%)
- ICSE 2015 Edward K. Smith, Christian Bird, and Thomas Zimmermann. Build it yourself! Home-grown Tools in a Large Software Company. In *Proceedings of the 37th International Conference on Software Engineering*, 2015, IEEE. (19%)
- ICSE 2015 Michael Barnett, Christian Bird, Joao Brunet, and Shuvendu K. Lahiri. Helping Developers Help Themselves: Automatic Decomposition of Code Review Changesets. In *Proceedings of the 37th International Conference on Software Engineering*, 2015, IEEE. (19%)
- MSR 2015 Christian Bird, Trevor Carnahan, and Michaela Greiler. Lessons Learned from Building and Deploying a Code Review Analytics Platform. In *Proceedings of the International Conference on Mining Software Repositories*, 2015, IEEE. (30%)
- MSR 2015 Amiangshu Bosu, Michaela Greiler, and Christian Bird. Characteristics of Useful Code Reviews: An Empirical Study at Microsoft. In *Proceedings of the International Conference on Mining Software Repositories*, 2015, IEEE. (30%)
- MSR 2015 Baishakhi Ray, Meiyappan Nagappan, Christian Bird, Nachiappan Nagappan, and Thomas Zimmermann. The Uniqueness of Changes: Characteristics and Applications. In *Proceedings of the International Conference on Mining Software Repositories*, 2015, IEEE. (30%)
- MSR 2015 Motahareh Bahrami Zanjani, Huzefa Kagdi, and Christian Bird. Using Developer-Interaction Trails to Triage Change Requests. In *Proceedings of the International Conference on Mining Software Repositories*, 2015, IEEE. (30%)
- TSE 2015 Emerson Murphy-Hill, Thomas Zimmermann, Christian Bird, and Nachiappan Nagappan. The Design Space of Bug Fixes and How Developers Navigate It. *IEEE Transactions on Software Engineering*, 2015
- FSE 2014 Miltiadis Allamanis, Earl T. Barr, Christian Bird, and Charles Sutton. Learning Natural Coding Conventions. In *Proceedings of the 22nd International Symposium on Foundations of Software Engineering*, 2014, ACM. (22%)
- ESEM 2014 Ashish Gupta, Thomas Zimmermann, Christian Bird, Nachiappan Nagappan, Thirumalesh Bhat, and Syed Emran. Mining Energy Traces to Aid in Software Development: An Empirical Case Study. In *Proceedings of the International Symposium on Empirical Software Engineering and Measurement*, 2014, ACM/IEEE



- ESE 2014 Abram Hindle, Christian Bird, Thomas Zimmermann, and Nachiappan Nagappan. Do topics make sense to managers and developers? *Empirical Software Engineering*, June 2014, 1–37
- ICSE 2014 Shaun Phillips, Thomas Zimmermann, and Christian Bird. Understanding and Improving Software Build Teams. In *Proceedings of the International Conference on Software Engineering*, 2014, IEEE. (20%)
- ICSE 2014 Kivanç Muşlu, Christian Bird, Nachiappan Nagappan, and Jacek Czerwinka. Transition from Centralized to Decentralized Version Control Systems: A Case Study on Reasons, Barriers, and Outcomes. In *Proceedings of the International Conference on Software Engineering*, 2014, IEEE. (20%)
- ICSE 2014 Christian Bird, Venkatesh Prasad Ranganath, Thomas Zimmermann, Nachiappan Nagappan, and Andreas Zeller. Extrinsic influence factors in software reliability: A study of 200,000 windows machines. In *Proceedings of the International Conference on Software Engineering (SEIP Track)*, 2014, IEEE. (21%)
- FSE 2013 Meiyappan Nagappan, Thomas Zimmermann, and Christian Bird. Diversity in software engineering research. In *Proceedings of the the joint meeting of the European Software Engineering Conference and the ACM SIGSOFT Symposium on The Foundations of Software Engineering (ESEC/FSE)*, 2013, ACM. (20%)
- FSE 2013 Peter C. Rigby, and Christian Bird. Convergent software peer review practices. In *Proceedings of the the joint meeting of the European Software Engineering Conference and the ACM SIGSOFT Symposium on The Foundations of Software Engineering (ESEC/FSE)*, 2013, ACM. (20%)
- ESEM 2013 Brendan Murphy, Christian Bird, Thomas Zimmermann, Laurie Williams, Nachiappan Nagappan, and Andrew Begel. Have Agile Techniques been the Silver Bullet for Software Development at Microsoft. In *Proceedings of the Seventh International Symposium on Empirical Software Engineering and Measurement*, 2013, ACM/IEEE. (28%)
- ISSTA 2013 Earl T. Barr, Christian Bird, and Mark Marron. Collecting a Heap of Shapes. In *Proceedings of the International Symposium on Software Testing and Analysis*, 2013, ACM. (25% Best Paper Award 2%)
- IEEE Software 2013 Robert Musson, Jacqueline Richards, Danyel Fisher, Christian Bird, Brian Bussone, and Sandipan Ganguly. Leveraging the Crowd: How 48,000 Users Helped Improve Lync Performance. *IEEE Software*, 2013
- ECSCW 2013 Richard Harper, Christian Bird, Thomas Zimmermann, and Brendan Murphy. Dwelling in Software: aspects of the felt-life of engineers in large software projects. In *Proceedings of the European Conference on Computer-Supported Cooperative Work*, 2013
- ICSE 2013 Alberto Bacchelli, and Christian Bird. Expectations, outcomes, and challenges of modern code review. In *Proceedings of the International Conference on Software Engineering*, 2013, IEEE. (18% nominated for Distinguished Paper 4%)

- ICSE 2013 Emerson Murphy-Hill, Thomas Zimmermann, Christian Bird, and Nachiappan Nagappan. The design of bug fixes. In *Proceedings of the International Conference on Software Engineering*, 2013, IEEE. (18%)
- ICSE 2013 Ekrem Kocaguneli, Thomas Zimmermann, Christian Bird, Nachiappan Nagappan, and Tim Menzies. Distributed development considered harmful? In *Proceedings of the International Conference on Software Engineering (Software Engineering in Practice Track)*, 2013, IEEE. (20%)
- RELENG 2013 Peter C. Rigby, Earl T. Barr, Christian Bird, Premkumar Devanbu, and Daniel M. German. What Effect does Distributed Version Control have on OSS Project Organization. In *Proceedings of the International Workshop on Release Engineering*, 2013, IEEE
- MSR 2013 Murtuza Mukadam, Christian Bird, and Peter C. Rigby. Gerrit software code review data from android. In *Proceedings of the International Working Conference on Mining Software Repositories (Data Track)*, 2013, IEEE
- CHASE 2013 Edward Smith, Robert Loftin, Emerson Murphy-Hill, Christian Bird, and Thomas Zimmermann. Improving Developer Participation Rates in Surveys. In *Proceedings of the International Workshop on Cooperative and Human Aspects of Software Engineering*, 2013, IEEE
- ESE 2012 Chris Parnin, Christian Bird, and Emerson Murphy-Hill. Adoption and Use of Java Generics. *Empirical Software Engineering*, 2012, 1–43
- FSE 2012 Christian Bird, and Thomas Zimmermann. Assessing the value of branches with what-if analysis. In *Proceedings of the 20th International Symposium on Foundations of Software Engineering*, 2012, ACM. (17% ACM Distinguished Paper Award)
- ICSM 2012 Abram Hindle, Christian Bird, Thomas Zimmermann, and Nachiappan Nagappan. Relating requirements to implementation via topic analysis: Do topics extracted from requirements make sense to managers and developers? In *Proceedings of the 28th IEEE International Conference on Software Maintenance*, 2012, IEEE. (25%)
- ESEM 2012 Emad Shihab, Christian Bird, and Thomas Zimmermann. The effect of branching strategies on software quality. In *Proceedings of the Sixth International Symposium on Empirical Software Engineering and Measurement*, 2012, ACM/IEEE. (25%)
- MSR 2012 Christian Bird, and Nachiappan Nagappan. Who? where? what? examining distributed development in two large open source projects. In *Proceedings of the International Working Conference on Mining Software Repositories*, 2012, IEEE. (28%)
- FASE 2012 Earl T. Barr, Christian Bird, Peter C. Rigby, Abram Hindle, Daniel M. German, and Premkumar Devanbu. Cohesive and Isolated Development with Branches. In *Proceedings of the International Conference on Fundamental Approaches to Software Engineering*, 2012, Springer. (24%)
- CSD 2012 Thomas Zimmermann, and Christian Bird. Collaborative Software Development in Ten Years: Diversity, Tools, and Remix Culture. In *Proceedings of the Workshop on The Future of Collaborative Software Development*, 2012

- ESE 2012 Foyzur Rahman, Christian Bird, and Premkumar Devanbu. Clones: What **is** that Smell? *Empirical Software Engineering, An International Journal*, 2012
- MLT 2011 Tim Menzies, Christian Bird, Tom Zimmermann, Wolfram Schulte, and Ekrem Kocaguneli. The Inductive Software Engineering Manifesto: Principles for Industrial Data Mining. In *Proceedings of the International Workshop on Machine Learning Technologies in Software Engineering*, 2011, ACM
- PROMISE 2011 Andreas Zeller, Thomas Zimmermann, and Christian Bird. Failure is a Four Letter Word: A Parody in Empirical Research. In *Proceedings of the 7th International Conference on Predictor Models in Software Engineering*, 2011
- ICSM 2011 Qiaona Hong, Sunghun Kim, S. C. Cheung, and Christian Bird. Understanding a Developer Social Network and its Evolution. In *Proceedings of the 27th IEEE International Conference on Software Maintenance*, 2011, IEEE. (28%)
- FSE 2011 Christian Bird, Nachiappan Nagappan, Brendan Murphy, Harald Gall, and Premkumar Devanbu. Don't Touch My Code! Examining the Effects of Ownership on Software Quality. In *Proceedings of the the eighth joint meeting of the European Software Engineering Conference and the ACM SIGSOFT Symposium on The Foundations of Software Engineering*, 2011, ACM. (17%)
- MSR 2011 Chris Parnin, Christian Bird, and Emerson Murphy-Hill. Java Generics Adoption: How New Features are Introduced, Champion, or Ignored. In *Proceedings of the International Working Conference on Mining Software Repositories*, 2011. (33%, invited to Journal special issue: 10%)
- CHASE 2011 Christian Bird, Thomas Zimmermann, and Alex Teterev. A Theory of Branches as Goals and Virtual Teams. In *Proceedings of the International Workshop on Cooperative and Human Aspects of Software Engineering*, 2011
- ESE 2010 Daryl Posnett, Christian Bird, and Premkumar Devanbu. An Empirical Study on the Influence of Pattern Roles on Change-Proneness. *Empirical Software Engineering, An International Journal*, 2010, 1–28
- FOSE 2010 Earl Barr, Christian Bird, Eric Hyatt, Tim Menzies, and Gregorio Robles. On the Shoulders of Giants. In *FSE/SDP Workshop on the Future of Software Engineering Research*, 2010. (65%)
- FSE 2010 Adrian Bachmann, Christian Bird, Foyzur Rahman, Premkumar Devanbu, and Abraham Bernstein. The Missing Links: Bugs and Bug-fix Commits. In *SIGSOFT '10/FSE-18: Proceedings of the 16th ACM SIGSOFT Symposium on Foundations of Software Engineering*, 2010, ACM. (20%)
- MSR 2010 Daryl Posnett, Christian Bird, and Premkumar Devanbu. Thex: Mining Metapatterns in Java. In *Proceedings of the Seventh Working Conference on Mining Software Repositories*, 2010, IEEE Computer Society. (31%)
- MSR 2010 Roozbeh Nia, Christian Bird, Premkumar Devanbu, and Vladimir Filkov. Validity of network analyses in open source projects. In *Proceedings of the Seventh Working Conference on Mining Software Repositories*, 2010, IEEE Computer Society. (31%)

- MSR 2010 Foyzur Rahman, Christian Bird, and Premkumar Devanbu. Clones: What is that Smell? In *Proceedings of the Seventh Working Conference on Mining Software Repositories*, 2010, IEEE Computer Society. (31% Best Paper Award 1%)
- ISSRE 2009 Christian Bird, Nachiappan Nagappan, Premkumar Devanbu, Harald Gall, and Brendan Murphy. Putting it All Together: Using Socio-Technical Networks to Predict Failures. In *Proceedings of the 17th International Symposium on Software Reliability Engineering*, 2009, IEEE Computer Society. (25%)
- FSE 2009 Christian Bird, Adrian Bachmann, Eirik Aune, John Duffy, Abraham Bernstein, Vladimir Filkov, and Premkumar Devanbu. Fair and Balanced? Bias in Bug-Fix Datasets. In *Proceedings of the the Seventh joint meeting of the European Software Engineering Conference and the ACM SIGSOFT Symposium on The Foundations of Software Engineering*, 2009, ACM. (14%)
- CACM 2009 Christian Bird, Nachiappan Nagappan, Premkumar Devanbu, Harald Gall, and Brendan Murphy. Does Distributed Development Affect Software Quality? An Empirical Case Study of Windows Vista. *Communications of the ACM* 52, 8, August 2009, 85–93. (Revised edition of ICSE paper invited to Research Highlights in CACM)
- ICSE 2009 Christian Bird, Nachiappan Nagappan, Premkumar Devanbu, Harald Gall, and Brendan Murphy. Does Distributed Development Affect Software Quality? An Empirical Case Study of Windows Vista. In *Proceedings of the 31st International Conference on Software Engineering*, 2009, IEEE Computer Society, pp. 518–528. (12% Best Paper Award 1%)
- MSR 2009 Christian Bird, Peter C. Rigby, Earl T. Barr, David J. Hamilton, Daniel M. German, and Prem Devanbu. The Promises and Perils of Mining Git. In *Proceedings of the Sixth Working Conference on Mining Software Repositories*, 2009, IEEE Computer Society. (28%)
- SDM 2009 Christian Bird, Earl Barr, Andre Nash, Premkumar Devanbu, Vladimir Filkov, and Zhendong Su. Structure and Dynamics of Research Collaboration in Computer Science. In *Proceedings of the Ninth SIAM International Conference on Data Mining*, 2009, SIAM, pp. 826–837. (29%)
- FSE 2008 Christian Bird, David Pattison, Raissa D’Souza, Vladimir Filkov, and Premkumar Devanbu. Latent Social Structure in Open Source Projects. In *SIGSOFT ’08/FSE-16: Proceedings of the 16th ACM SIGSOFT Symposium on Foundations of Software Engineering*, 2008, ACM, pp. 24–35. (20%)
- MSR 2008 David Pattison, Christian Bird, and Premkumar Devanbu. Talk and Work: a Preliminary Report. In *Proceedings of the Fifth International Working Conference on Mining Software Repositories*, 2008, ACM, pp. 113–116. (40%)
- FSE 2007 Zachary M. Saul, Vladimir Filkov, Premkumar Devanbu, and Christian Bird. Recommending Random Walks. In *ESEC-FSE ’07: Proceedings of the the Sixth joint meeting of the European Software Engineering Conference and the ACM SIGSOFT Symposium on The Foundations of Software Engineering*, 2007, ACM, pp. 15–24. (17%, nominated for Best Paper)

- MSR 2007 Christian Bird, Alex Gourley, and Prem Devanbu. Detecting Patch Submission and Acceptance in OSS Projects. In *Proceedings of the Fourth International Workshop on Mining Software Repositories*, 2007, IEEE Computer Society, pp. 26–35. (38%)
- MSR 2007 Christian Bird, Alex Gourley, Prem Devanbu, Anand Swaminathan, and Greta Hsu. Open Borders? Immigration in Open Source Projects. In *Proceedings of the Fourth International Workshop on Mining Software Repositories*, 2007, IEEE Computer Society, p. 6. (38%)
- APVIS 2007 Michael Ogawa, Kwan-Liu Ma, Christian Bird, Premkumar T. Devanbu, and Alex Gourley. Visualizing Social Interaction in Open Source Software Projects. In *Sixth International Asia-Pacific Symposium on Visualization*, 2007, pp. 25–32. (45%)
- MSR 2006 Christian Bird, Alex Gourley, Prem Devanbu, Michael Gertz, and Anand Swaminathan. Mining Email Social Networks. In *Proceedings of the Third International Workshop on Mining software repositories*, 2006, ACM, pp. 137–143. (35%)
- MSR 2006 Christian Bird, Alex Gourley, Prem Devanbu, Michael Gertz, and Anand Swaminathan. Mining Email Social Networks in Postgres. In *Proceedings of the Third International Workshop on Mining software repositories (Challenge Track)*, 2006, ACM, pp. 185–186

### Invited Papers, Demo papers, etc.

- IEEE Software 2018 P. Devanbu, T. Zimmermann, and C. Bird. Belief and Evidence: How Software Engineers Form Their Opinions. *IEEE Software* 35, 6, November/December 2018, 72–76
- ICSM 2011 Christian Bird. Sociotechnical Coordination and Collaboration in Open Source Software. In *Doctoral Symposium, Proceedings of the 27th IEEE International Conference on Software Maintenance*, 2011, IEEE
- CSCW 2011 Christian Bird, Brendan Murphy, Nachiappan Nagappan, and Thomas Zimmermann. Empirical Software Engineering at Microsoft Research. In *Showcase Track, Proceedings of the ACM Conference on Computer Supported Cooperative Work*, 2011
- ESEM 2010 Foyzur Rahman, Christian Bird, and Premkumar Devanbu. Clones: What is that Smell? In *Proceedings of the fourth International Symposium on Empirical Software Engineering and Measurement*, 2010, IEEE Computer Society. (invited paper)
- FSE 2010 Christian Bird, Adrian Bachmann, Foyzur Rahman, and Abraham Bernstein. Linkster: Enabling Efficient Manual Mining. In *Demonstration Track, Proceedings of the 17th SIGSOFT Symposium on Foundations of Software Engineering*, 2010, ACM. (accepted for formal demonstration, 21%)

### Invited Talks

- 2020 **Lessons and Insights from Tech Transfers at Microsoft**, Keynote, *The 27th IEEE International Conference on Software Analysis, Evolution and Reengineering*, London, Ontario Canada
- 2019 **Lessons and Insights from Tech Transfers at Microsoft**, Keynote, *The 10th Brazilian Conference on Software*, Salvador, Bahia, Brazil

- 2019 **Lessons and Insights from Tech Transfers at Microsoft**, *Keynote, The 15th International Conference on Predictive Models and Data Analytics in Software Engineering*, Porto de Galinhas, Brazil
- 2014 **Lessons and Insights from Tech Transfers at Microsoft**, *Keynote, 4th Workshop on Mining Unstructured Data (colocated with ICSME 2014)*, Victoria, British Columbia, Canada
- 2014 **Empirical Investigations into Code Review at Microsoft and Beyond**, *Carnegie Mellon University*, Pittsburgh, Pennsylvania
- 2013 **Investigations into Code Review at Microsoft and Beyond**, *University of Victoria*, Victoria, British Columbia, Canada
- 2012 **Branch Analytics at Microsoft**, *University of Victoria*, Victoria, British Columbia, Canada
- 2011 **Empirical Software Engineering at Microsoft Research: Transitioning Research into practice**, *Brigham Young University*, Provo, Utah
- 2011 **SocioTechnical Effects in Software Teams**, *Hong Kong University of Science and Technology*, Hong Kong
- 2010 **SocioTechnical Effects in Software Teams**, *Microsoft Research*, Bangalore, India
- 2010 **Does Distributed Development Affect Software Quality?**, *Third India Software Engineering Conference*, Mysore, India
- 2010 **Jazz Anthropology**, *T. J. Watson Research Center, IBM*, Hawthorne, New York
- 2009 **On the Effect of Ownership in Varying Software Processes**, *Microsoft Research*, Redmond, Washington
- 2008 **Latent Social Structure in Open Source Projects**, *INFORMS Annual Meeting*, Washington, D.C..
- 2008 **Does Distributed Development Affect Software Quality?**, *Microsoft Research*, Redmond, Washington
- 2008 **Social Organization in Open Source Software**, *T. J. Watson Research Center, IBM*, Hawthorne, New York, U.S.A.
- 2006 **Social Network Analysis in Software Teams**, *Guest Lecture, MAE 298, Understanding Networks: Theory and Applications*, University of California, Davis, U.S.A.

## Research Experience

- 2022 - present **Senior Principal Researcher**, *Empirical Software Engineering Group*, Microsoft, Research, Redmond, Washington
- 2017 - 2022 **Principal Researcher**, *Empirical Software Engineering Group*, Microsoft, Research, Redmond, Washington
- 2013 - 2017 **Senior Researcher**, *Empirical Software Engineering Group*, Microsoft, Research, Redmond, Washington

- 2011 - 2013 **Researcher**, *Empirical Software Engineering Group*, Microsoft, Research, Redmond, Washington
- 2011 - **Senior Researcher**, *Empirical Software Engineering Group*, Microsoft, Research, Redmond, Washington
- 2010 - 2011 **Postdoctoral Researcher**, *Empirical Software Engineering Group*, Microsoft, Research, Redmond, Washington
- 2005-2010 **Research Assistant**, *Software Engineering Lab under Prof. Premkumar Devanbu*, University of California, Davis, Davis, California
- 2009 **Research Intern**, *Governance Science Research Group under Clay Williams*, T. J. Watson Research Center, IBM, Hawthorne, New York
- 2009 **Research Intern**, *Software Engineering Group under Nachiappan Nagappan*, Microsoft Research, Redmond, Washington
- 2008 **Research Intern**, *Software Engineering Group under Nachiappan Nagappan*, Microsoft Research, Redmond, Washington
- 2004 **Research Assistant**, *Software Engineering Lab under Prof. Raju Pandey*, University of California, Davis, Davis, California

## Teaching Experience

- 2004 - 2005 **Teaching Assistant**, *Programming Languages under Prof. Ronald Olsson and Prof. Raju Pandey*, University of California, Davis, Davis, California
- 1997 **Teaching Assistant**, *Advanced Data Structures under Prof. Theodore Norman*, Brigham Young University, Provo, Utah

## Industry Experience

- 2000-2005 **Software Engineer**, *Embedded Software Tools Group*, Motorola, Lindon, Utah
- 1997 **Technical Support Engineer**, *Technical Support Department*, Caldera Systems Inc., Provo, Utah

## Academic Service

- PC Chair** International Conference on Automated Software Engineering, 2023
- International Conference on Automated Software Engineering, Demonstrations Track, 2022
- International Conference on Software Analysis, Evolution and Reengineering, Industry Track, 2020
- ACM Sigsoft Symposium on Foundations of Software Engineering, Journal First Track, 2020
- International Symposium on Empirical Software Engineering and Measurement, Industry Track, 2018

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|                     | International Conference on Software Maintenance and Evolution, Industry Track, 2018                                                  |
|                     | International Conference on Mining Software Repositories, 2016                                                                        |
|                     | International Conference on Program Comprehension, 2015                                                                               |
|                     | International Workshop on Replications in Software Engineering, 2013                                                                  |
|                     | International Workshop on Data Analysis Patterns in Software Engineering (DAPSE), 2013                                                |
|                     | International Workshop on Release Engineering (RelEng), 2013                                                                          |
|                     | International Conference on Program Comprehension, Industry Track, 2013                                                               |
|                     | Challenge Track of Mining Software Repositories, 2009                                                                                 |
| <b>Journal</b>      | Review Board, IEEE Transactions on Software Engineering, 2017-present                                                                 |
| <b>Positions</b>    | Associate Editor, Journal of Systems and Software, 2017-2018                                                                          |
|                     | Academic Editor, PeerJ Computer Science, 2016-present                                                                                 |
|                     | Editorial Board, Journal of Information and Software Technology, 2016-present                                                         |
| <b>Guest Editor</b> | Journal of Empirical Software Engineering Special Issue for International Conference on Program Comprehension 2015, published in 2017 |
|                     | Journal of Empirical Software Engineering Special Issue for Mining Software Repositories 2016, in progress                            |
|                     | IEEE Software Special Issue on Crowdsourcing in Software Engineering, 2017                                                            |
|                     | IEEE Software Special Issue on Release Engineering, 2015                                                                              |
|                     | IEEE Software 2nd Special Issue on Release Engineering, 2016                                                                          |
|                     | IEEE Software 3rd Special Issue on Release Engineering, 2017                                                                          |
| <b>Steering</b>     | International Conference on Mining Software Repositories MSR, 2015-present                                                            |
| <b>Committee</b>    | International Conference on Program Comprehension (ICPC), 2015-2018                                                                   |
| <b>Organizing</b>   | International Workshop on Data Analysis Patterns in Software Engineering (DAPSE), 2013                                                |
| <b>Committee</b>    | International Workshop on Release Engineering (RelEng), 2013, 2014, 2015, 2016                                                        |
|                     | International Workshop on Replications in Software Engineering, 2013                                                                  |
| <b>PC Member</b>    | International Conference on Software Engineering, 2014, 2016, 2017, 2019, 2020, 2022                                                  |
|                     | ACM Sigsoft Symposium on Foundations of Software Engineering, 2012, 2016, 2018, 2022                                                  |
|                     | International Symposium on Software Testing and Analysis, 2019                                                                        |
|                     | International Conference on Automated Software Engineering, 2014, 2015, 2018, 2020, 2022, 2023                                        |



International Symposium on Empirical Software Engineering and Measurement, 2015, 2017

Indian Software Engineering Conference, 2015, 2016

International Symposium on Software Reliability Engineering, Industrial track, 2013

International Conference on Software Maintenance and Engineering, 2012, 2013, 2014, 2022

International Conference on Program Comprehension, 2012, 2014, 2016, 2018, 2022

International Conference on Predictor Models in Software Engineering, 2013

International Conference on Software Engineering Advances, 2013

International Conference on Mining Software Repositories, 2010, 2011, 2012, 2013, 2015, 2017, 2018, 2022

International Conference on Global Software Engineering, 2015, 2016

International Conference on Software Engineering and Applications, 2014

International Conference on Software Analysis, Evolution and Reengineering, 2015, 2016, 2017, 2018, 2019, 2020

International Conference on Software Maintenance and Evolution, Early Research Achievements track, 2016

International Symposium on Software Crowdsourcing, 2015, 2016

ACM Student Research Competition (at ICSE), 2012

Formal Research Demonstrations track at ICSE, 2014

ACM Sigsoft Symposium on Foundations of Software Engineering, Tool Demonstration Track, 2017

International Conference on Software Analysis, Evolution, and Reengineering, Early Research Achievements track, 2016

International Conference on Software Analysis, Evolution, and Reengineering, Reproducibility and Negative Results track, 2018

International Symposium on Empirical Software Engineering and Measurement, Industrial track, 2011

International Conference on Software Maintenance, Early Achievements track, 2011

European Soft. Eng. Conf./Foundations of Soft. Eng., Artifact Evaluation Committee, 2011

Euromicro Conference on Software Engineering and Advanced Applications Special Session on Estimation and Prediction in Software & Systems Engineering, 2012, 2014

2nd SEMAT Workshop on a General Theory of Software Engineering (GTSE) 2013,

International Workshop on Software Refactoring, 2016

International Workshop on Alternative Workforces for Software Engineering 2015

International Workshop on Regression Testing, 2012, 2013

International Workshop on Machine Learning Technologies in Software Engineering, 2011

International Workshop on Empirical Software Engineering in Practice, 2010, 2011, 2012, 2013

International Workshop on Replications in Empirical Software Engineering Research, 2010, 2011

International Workshop on Emerging Trends in FLOSS Research and Development, 2010

International Workshop on the General Theory of Software Engineering, 2013

Challenge Track of Mining Software Repositories, 2008, 2009

Data Track of Mining Software Repositories, 2014

**Reviewer** Communications of the ACM

ACM Transactions on Software Engineering and Methodology

IEEE Transactions on Software Engineering

Journal of Empirical Software Engineering

IEEE Software

IEEE Intelligent Systems

Journal of Systems and Software

Journal of Information and Software Technology

Journal of Software: Evolution and Process

Journal of Software Testing, Verification & Reliability

Software Quality Journal

International Symposium on Software Testing and Analysis, 2012

International Conference on Software Engineering, 2008, 2009, 2013

International Conference on Software Engineering (SEIP Track), 2015

ACM Conference on Computer Supported Cooperative Work, 2008, 2012, 2013

International Working Conference on Mining Software Repositories, 2007, 2008, 2009

European Conference on Object Oriented Programming, 2009

Haifa Verification conference, 2011

ACM Conference on Recommender Systems, 2009

Science of Computer Programming, 2011

**Other** Student Volunteer Chair, Sigsoft Symposium on Foundations of Software Engineering, 2014

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## Thesis Committees

Katherine Dearstyne, Ph.D., Notre Dame University, 2025

Farhshad Kazemi, Ph.D., University of Waterloo, 2024

Emad Aghayi, Ph.D., George Mason University, 2020

Motahareh (Sara) Bahrami Zanjani, Ph.D., Wichita State University, 2017

Jordan Ell, M.S., University of Victoria, 2014

Adrian Schröter, Ph.D., University of Victoria, 2012

## Memberships

Senior Member, Institute of Electrical and Electronics Engineers (IEEE)

Distinguished Scientist, Association for Computing Machinery (ACM)

ACM Special Interest Group on Software Engineering (ACM SIGSOFT)

## References

Available upon request